Flavio Giuliani

Ph.D. in Physics

Current address & contacts

Dipartimento di Fisica, Sapienza University of Rome
P.le A. Moro 5, 00185 - Rome, Italy

- My profile on scientific databases: Google Scholar, ResearchGate, ORCID
- My GitHub profile and my personal website: github.com/flagiu flagiu.github.io

Education

- 2022 - 2026: **Ph.D. in Physics**, XXXVIII ciclo, ottimo con lode (maximum grade with honors), Sapienza University of Rome.
  - Thesis: *Study of the thermodynamics and kinetic anomalies of Antimony-based phase-change materials by machine-learned molecular dynamics*, supervised by Prof. Riccardo Mazzarello and Prof. Lilia Boeri
    - The thesis project involved the study of thermodynamic and kinetic anomalies and crystallization features in supercooled liquid Sb, a candidate phase-change material, and the development of a neural network interatomic potential trained on density functional theory for Ge-alloyed Sb
- 2020 - 2022: **Master's Degree in Physics**, 110/110 e lode (maximum grade with honors), Sapienza University of Rome.
  - Thesis: *Vibrational dynamics of simulated glasses with different fictive temperatures*, supervised by Prof. Tullio Scopigno and Prof. Francesco Sciortino
    - The aim of the thesis project was to assess the correlation between supercooled liquid viscosity and glass vibrational properties for a prototype fragile glass former: the rigid model of ortho-terphenyl
- 2017 - 2020: **Bachelor's Degree in Physics**, 110/110 e lode (maximum grade with honors), Sapienza University of Rome
  - Thesis: *Twitching motility in bacteria*, supervised by Prof. Roberto Di Leonardo
    - The thesis project consisted in reproducing a tug-of-war physical model for simulating the twitching dynamics of a bacterium on a surface, mediated by the adhesion and retraction of type-IV pili

Publications

I have published 2 articles in international peer-reviewed scientific journals. I am the first author of 1 of those publications, on the high-Impact Factor (IF) journal "Proceedings of the National Academy of Sciences U.S.A." (PNAS U.S.A., IF: 9.1).

1. F. Giuliani, F. G. Mattioli, Y. Chen, D. Baratella, D. Dragoni, M. Bernasconi, J. Russo, L. Boeri & R. Mazzarello, "Liquid anomalies and fragility of supercooled antimony", PNAS U.S.A. 123 (12) e2531605123, DOI: 10.1073/pnas.2531605123 (2026) [article][preprint]
2. S. Ritarossi, R. Piombo, F. Giuliani, D. Dragoni, M. Bernasconi, R. Mazzarello, "Phase-Change Heterostructures Based on Antimony", *Phys. Status Solidi RRL* **19** (7), 2500012 (2025) [article]

Teaching

- 2024: **Teaching Assistant** for "Mathematics and Statistics", Bachelor's Degree in Environmental Sciences, Sapienza University of Rome. Interactive exercises and homework corrections (40 h)
- 2023: 1 h lecture "Machine learning interatomic potentials" for the course "Computing Methods for Physics" of Master's Degree in Physics by Prof. L. Boeri, Sapienza University of Rome

- 2021, 2022: **Student-collaboration scholarship** for Scientific Programming in C ("Laboratorio di Calcolo"), Bachelor's Degree in Physics, Sapienza University of Rome. Laboratory assistant and notes-writer (75 h/year)

Certified advanced and soft-skill courses

- 2024: Course "HPC Molecular Modelling", three-days lessons and hands-on at CINECA, Rome
- 2022 - 2023: Within my Ph.D. education, Sapienza University of Rome
  - "Statistical Field Theory" (40 h, 6 credits) by Dr. A. Cavagna, 30/30 e lode (maximum grade with honors)
  - "Advanced Topics in Condensed Matter" (40 h, 6 credits) by Prof. J. Lorenzana and Dr. P. Barone, 30/30 con lode (maximum grade with honors)
  - Soft-skills course "Preparing artwork for scientific papers" by Prof. G. Giardina
  - Soft-skills online courses "I principali programmi europei di finanziamento R&I", "Etica e Scienza", "Fare start-up in Sapienza, istruzioni per l'uso" and "Publications and research data: Open Access"

Grants

High-Performance Computing (HPC) resources

- 2025: **20k GPUh grant** as Principal Investigator (PI) of the ISCRA-C project "LIGAN: LIquid-liquid transition in Germanium-doped Antimony with a Neural-network interaction potential" for CINECA's Leonardo supercomputer
- 2024: **20k GPUh grant** as PI of the ISCRA-C project "AntimoNN: Neural-Network interaction potential for Antimony-based phase-change materials" for CINECA's Leonardo supercomputer
- 2023: **23333 CPUh grant** as PI of the ISCRA-C project "DAGAN: First-principles DAtaset of Germanium-Antimony for a Neural-network interaction potential" for CINECA's G100 supercomputer

Other grants

- 2025: **€11k grant for the University call for Starting Third Mission initiatives**, Sapienza University of Rome
  - Title: *Eutopie - Dialogues on the relationship between science and society for a research on human beings*
  - Role: Proponent and Head of Organization. I oversaw the entire organization cycle: format design, selection of interdisciplinary topics, research and involvement of 15+ high-profile experts, logistics management, communication, and feedback analysis through questionnaires.
  - Objective and implementation: The project created an open space for multidisciplinary debate in the university to address major contemporary social challenges, promoting collective involvement through four public dialogues on: 1) Climate crisis and human-nature relationship; 2) Science, women, and rights; 3) Human language and artificial intelligence; 4) Research and imagination in art and science.

Awards

- 2022: Excellence program scholarship of Master's Degree in Physics, Sapienza University of Rome
  - Awarded to top 15 students based on exam marks
- 2020: Excellence program scolarship of Bachelor's Degree in Physics, Sapienza University of Rome
  - Awarded to top 34 students based on exam marks

Conferences & seminars

Contributions

- 2024: **Poster** presentation at "CMT@BRIXEN '24: The Meeting of the Condensed Matter Theory Italian Community", Brixen, Italy.
  - Poster: Ab initio study of supercooled antimony alloys through a machine learning interatomic potential

- 2024: **10’ talk** at the worskshop “Multiscale Modeling and Engineering Applications”, ICSC - National Research Center in HPC, Big Data and Quantum Computing - Spoke 6, Sapienza University of Rome
  - Talk: “Neural-network interatomic potential for antimony-based phase-change materials”, within the Flagship Project FP1 - Advanced Materials and New Devices
- 2023: **20’ talk** at the PhD Journal club seminar of the Physics Department, Sapienza University of Rome
  - Talk: “Accessing large-scale molecular simulations through machine learning”

Participant

- 2024: Participant at “European Phase-Change and Ovonics Symposium” (E\PCOS2024), Leibniz Institute of Surface Engineering (IOM), Leipzig, Germany
- 2023: Participant and Technical staff at E\PCOS2023, Sapienza University of Rome
- 2022: Participant at Workshop “From Water to Colloidal Water - Celebrating the contributions of Francesco Sciortino to Soft Matter”, Sapienza University of Rome
- 2021: Participant at “XXVII INC Summer School: Ultrastable Glasses - New perspectives for an old problem”, organized by Instituto Nicolás Cabrera, Universidad Autónoma de Madrid, in “La Cristalera”, Miraflores de la Sierra, Madrid, Spain
- 2020: Participant at “Summer School: Physics of Life”, organized by the Center for the Physics of Biological Function, Princeton University, online

Scientific communication for a broad audience

- 2025: Talk for the book presentation "L’atomica e le responsabilità della scienza, organized by L’Asino d’oro Edizioni at Libreria Spazio Sette, Rome

Volunteering

- 2019: Co-founder of “STudy to transfORM” (ST.ORM) (website, Instagram), a nonprofit organization whose educational projects currently span across Italy, Kenya and Tanzania
- 2019: For about one year, I taught math and sciences to refugees within the activities of the non-profit organization “Baobab Experience” in Rome

Personal skills

- Language: Italian (native), English (B2), Spanish (A2)
- Driving license: Italian B